

An introduction to Bayesian modelling with brms and Stan

Francisco Rodríguez-Sánchez

<https://frodriguezsanchez.net>

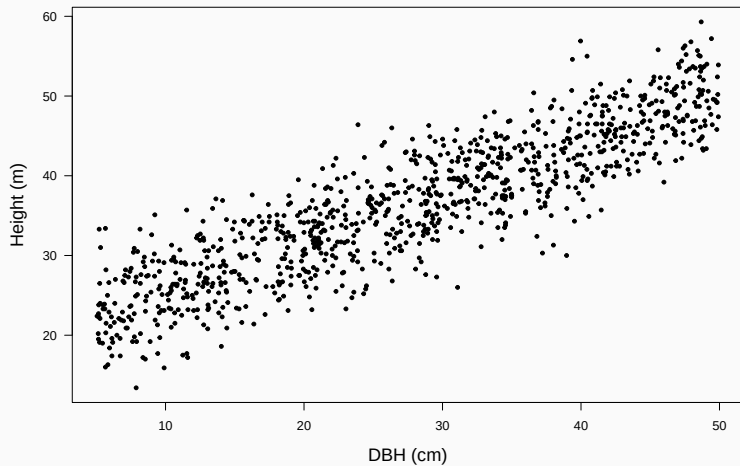
Our dataset: tree heights and DBH

- One species
- 10 plots
- 1000 trees
- Number of trees per plot ranging from 4 to 392

```
trees <- read.csv('data/trees.csv')
```

	site	dbh	height
Min.	: 1.0	Min. : 5.06	Min. :13.40
1st Qu.:	1.0	1st Qu.:17.69	1st Qu.:29.68
Median	: 2.0	Median :28.62	Median :36.55
Mean	: 2.7	Mean :27.88	Mean :36.51
3rd Qu.:	4.0	3rd Qu.:38.97	3rd Qu.:43.33
Max.	:10.0	Max. :49.92	Max. :59.30

What's the relationship between DBH and height?



First step: linear regression (lm)

```
simple.lm <- lm(height ~ dbh, data = trees)
```

Call:

```
lm(formula = height ~ dbh, data = trees)
```

Residuals:

Min	1Q	Median	3Q	Max
-13.3270	-2.8978	0.1057	2.7924	12.9511

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	19.33920	0.31064	62.26	<2e-16 ***
dbh	0.61570	0.01013	60.79	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.093 on 998 degrees of freedom

Multiple R-squared: 0.7874, Adjusted R-squared: 0.7871

F-statistic: 3695 on 1 and 998 DF, p-value: < 2.2e-16

Center continuous variables

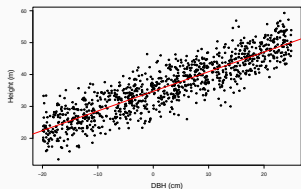
```
summary(trees$dbh)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
5.06	17.69	28.62	27.88	38.97	49.92

```
trees$dbh.c <- trees$dbh - 25
```

So, all parameters will be referred to a 25 cm DBH tree.

Linear regression with centred DBH



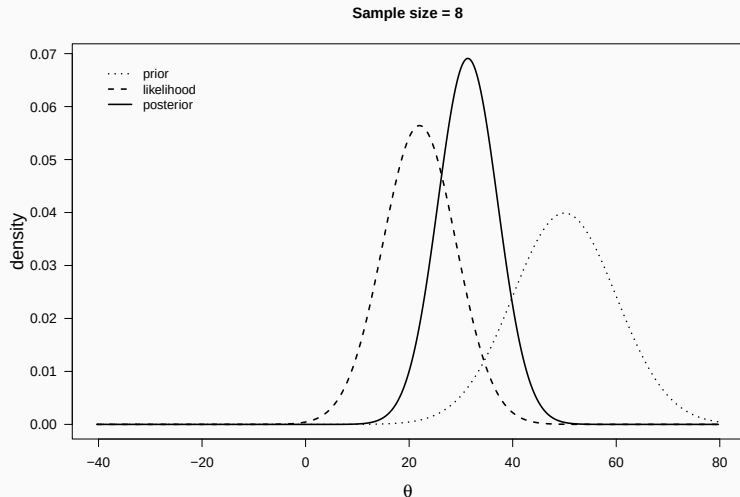
```
lm(formula = height ~ dbh.c, data = tree)
      coef.est coef.se
(Intercept)  34.73    0.13
dbh.c         0.62    0.01
---
n = 1000, k = 2
residual sd = 4.09, R-Squared = 0.79
```

Let's make it Bayesian

Bayesian inference: prior, posterior, and likelihood

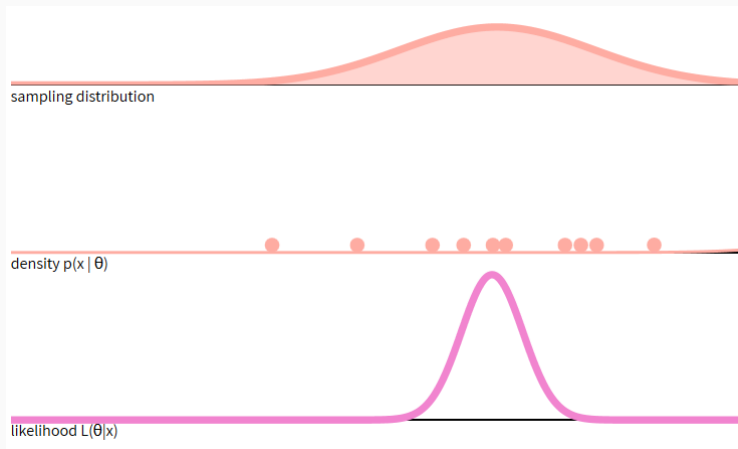
$$P(\text{Unknown}|\text{Data}) \propto P(\text{Data}|\text{Unknown}) \times P(\text{Unknown})$$

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$



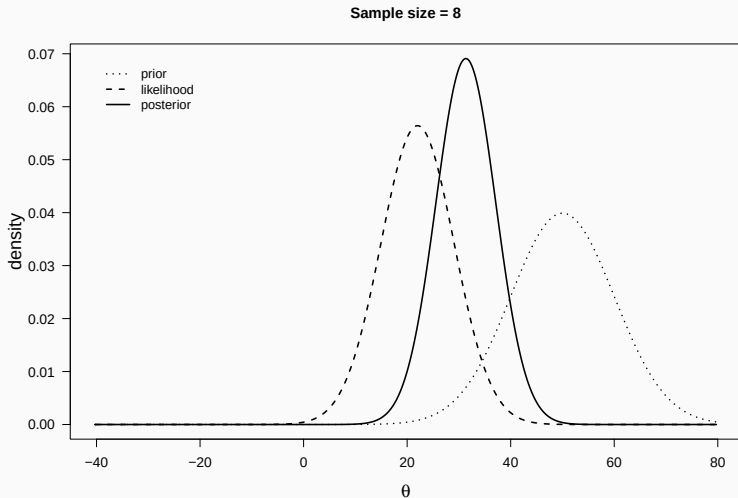
What is the likelihood?

$$L(\theta|x) = P(x|\theta)$$

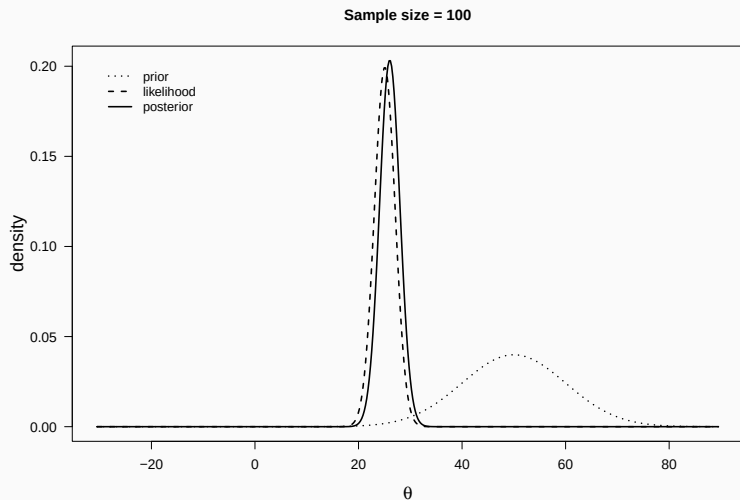


<https://seeing-theory.brown.edu/bayesian-inference/index.html>

Bayesian inference: prior and likelihood produce posterior



With increasing sample size, likelihood dominates prior



- Integrate information (**prior**)

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- Particularly important with **limited** sample sizes
- Large dataset -> prior effect **diminishes**
- **Uncertainty / Propagate errors**

$$y_i \sim N(\mu_i, \sigma^2)$$

$$\mu_i = \alpha + \beta x_i$$

In this case:

$$\text{Height}_i \sim N(\mu_i, \sigma^2)$$

$$\mu_i = \alpha + \beta \text{DBH}_i$$

α : expected height when DBH = 25 cm

β : how much height increases with every unit increase of DBH

```
library('brms')  
  
height.formu <- brmsformula(height ~ dbh.c)
```

We must define **prior distributions** for every parameter

```
get_prior(height.formu, data = trees)
```

	prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag
	(flat)	b								
	(flat)	b	dbh.c							
student_t(3, 36.5, 10.2)	Intercept									
student_t(3, 0, 10.2)	sigma							0		
source										
default										
(vectorized)										
default										
default										

Choosing priors

Avoid 'non-informative' priors

Use *weakly informative* (e.g. relatively wide Normal or t-student distributions)
or *strongly informative* priors based on previous knowledge and common sense.

Some tips for setting priors:

- <https://github.com/stan-dev/stan/wiki/Prior-Choice-Recommendations>

Run **prior predictive checks** (just priors, no data)

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- [Priors chapter](#) in The BUGS book

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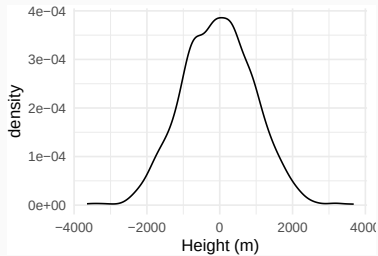
- <https://github.com/stan-dev/stan/wiki/Prior-Choice-Recommendations>
- Priors chapter in The BUGS book
- <https://doi.org/10.1111/oik.05985>

Run **prior predictive checks** (just priors, no data)

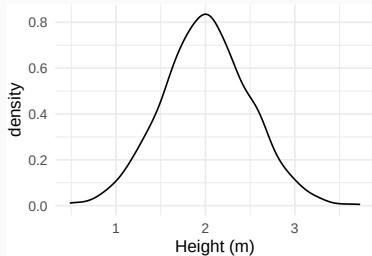
<https://distribution-explorer.github.io/>

Example: estimating people height across countries

Unreasonable prior



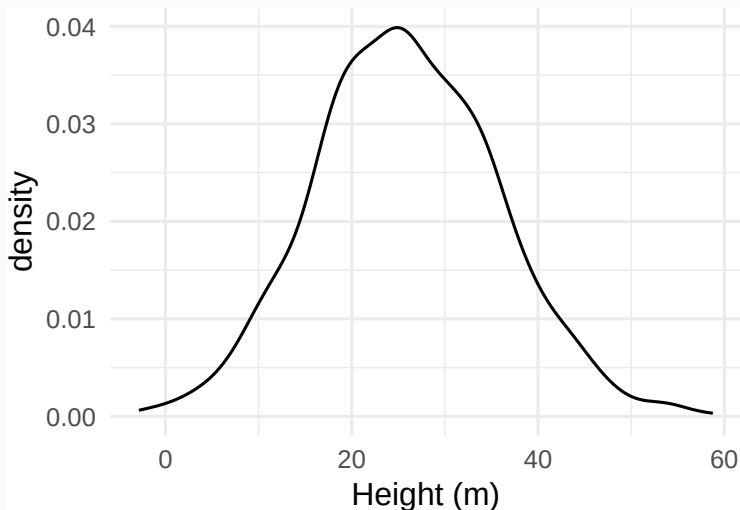
Reasonable prior



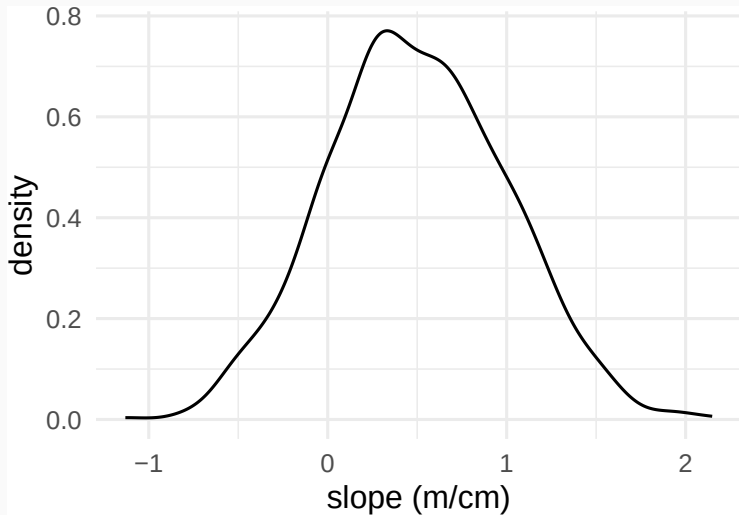
Defining priors for our trees example

```
priors <- c(  
  set_prior('normal(30, 10)', class = 'Intercept'),  
  set_prior('normal(0.5, 0.4)', class = 'b'),  
  set_prior('normal(0, 5)', class = 'sigma')  
)
```

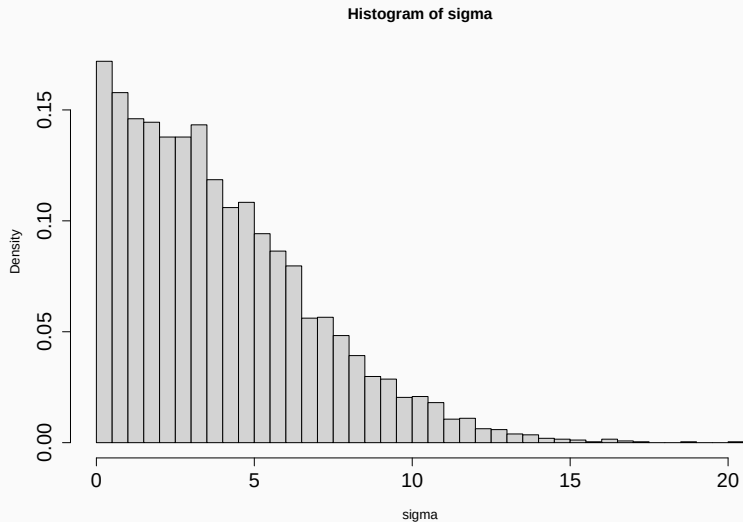
Prior for intercept (average height of 25-cm diameter tree)



Prior for slope



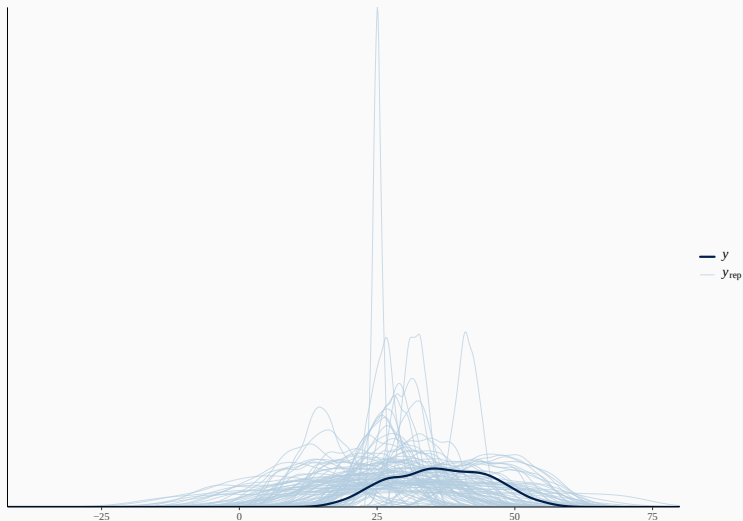
Prior for sigma (residual sd)



```
height.mod <- brm(height.formu,  
  data = trees,  
  prior = priors,  
  sample_prior = 'only')
```

Prior predictive check

```
pp_check(height.mod, ndraws = 100)
```



```
height.mod <- brm(height.formu,  
  data = trees,  
  prior = priors)
```


Model summary

```
summary(height.mod)
```

```
Family: gaussian
Links: mu = identity
Formula: height ~ dbh.c
Data: trees (Number of observations: 1000)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	34.73	0.13	34.47	34.99	1.00	4222	3247
dbh.c	0.62	0.01	0.60	0.64	1.00	4499	2673

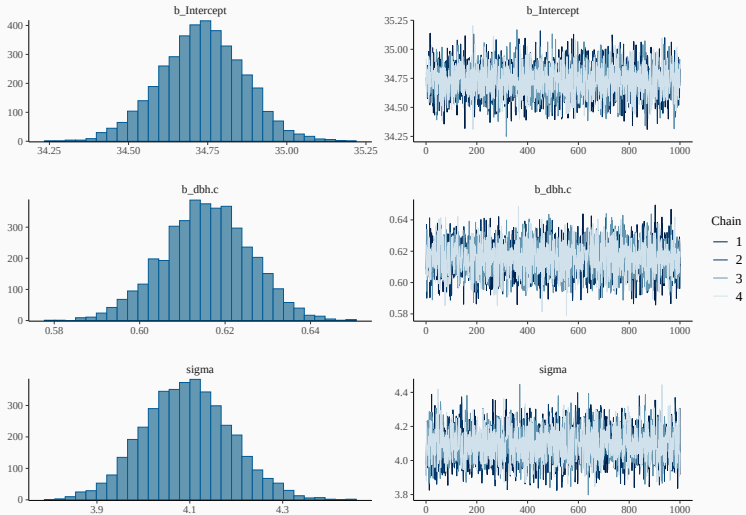
Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	4.09	0.09	3.92	4.28	1.00	4430	3153

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

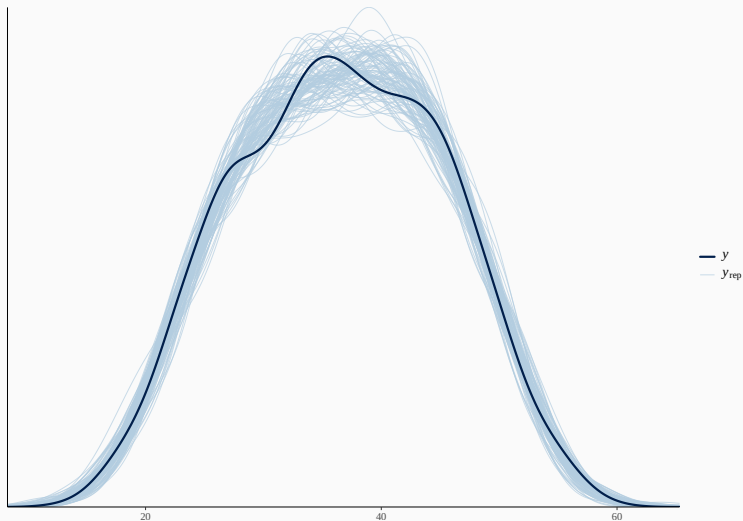
Model visualisation

```
plot(height.mod)
```



Posterior predictive checking

```
pp_check(height.mod, ndraws = 100)
```



```
library('shinystan')  
launch_shinystan(height.mod)
```

Sensitivity based on cjs_dist

Prior selection: all priors

Likelihood selection: all data

variable	prior	likelihood	diagnosis
b_Intercept	0.001	0.084	-
b_dbh.c	0.001	0.087	-
sigma	0.003	0.100	-
Intercept	0.001	0.085	-

Prior sensitivity analysis

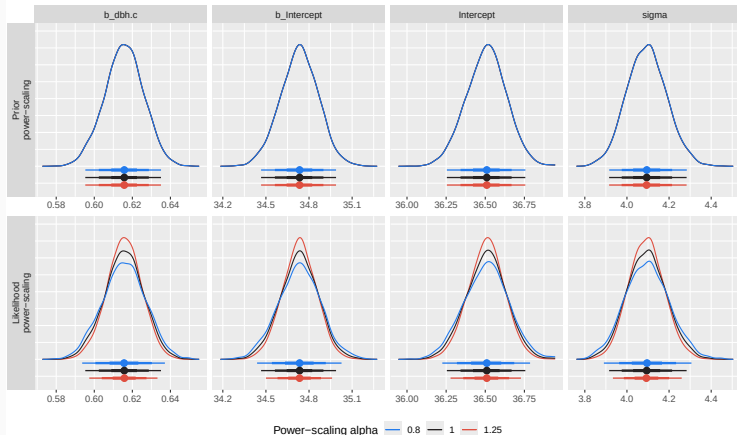
Power-scaling sensitivity

Posterior density estimates depending on amount of power-scaling (α).

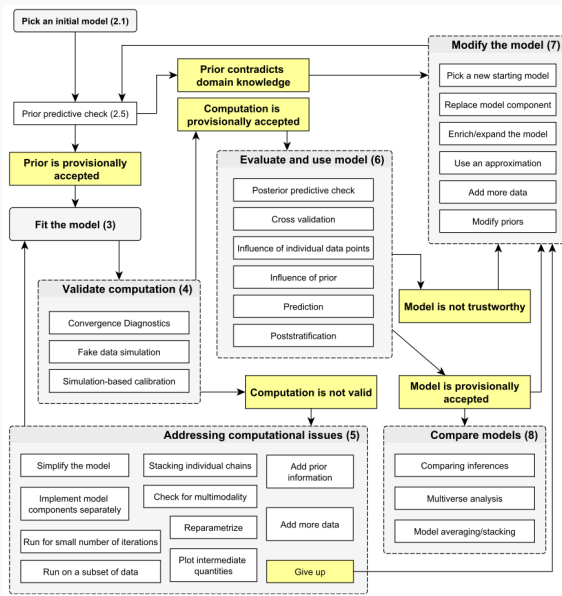
Overlapping lines indicate low sensitivity.

Wider gaps between lines indicate greater sensitivity.

Estimates with high Pareto k (dashed lines) may be inaccurate.



The Bayesian workflow



height ~ sex

[Regression and other stories](#)

[Statistical Rethinking](#)

[Statistical rethinking with brms, ggplot2, and the tidyverse](#)

[Bayesian Population Analysis using WinBugs](#)

[Applied Hierarchical Modeling in Ecology](#)

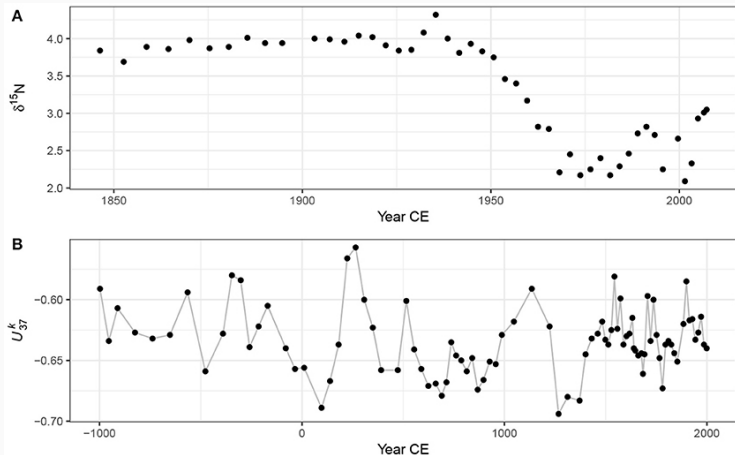
[Stan user guide](#)

Generalised Additive Models

Francisco Rodríguez-Sánchez

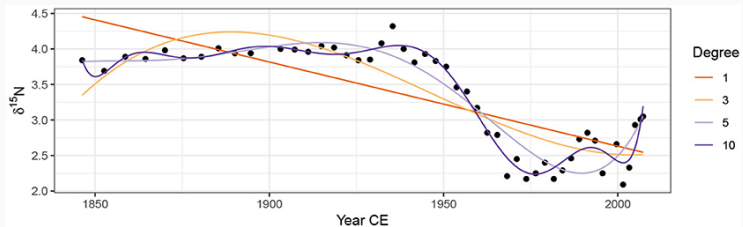
<https://frodriguezsanchez.net>

How do we model these time series?



Simpson 2018

How do we model these time series?



Simpson 2018

Generalised Linear Model (GLM):

$$y = a + bx$$

Generalised Additive Model (GAM):

$$y = a + s(x)$$

Modelling non-linear time series with GAM

```
isotopes <- readRDS('data/isotope.rds')
```

	Depth	d13C	TotalC	d15N	TotalN	DryWeight	Year
1	0.2	-27.57	806.49	3.05	64.21	8.2	2007.254
2	0.4	-27.67	949.33	3.01	73.26	7.6	2006.510
3	0.8	-27.63	1305.52	2.93	93.25	11.6	2004.941
4	1.2	-27.62	1136.04	2.33	86.09	9.6	2003.269
5	1.6	-27.48	1028.27	2.09	93.80	10.9	2001.496
6	2.0	-27.39	809.91	2.66	79.98	9.9	1999.626

Modelling non-linear time series with GAM

```
library('mgcv')  
m <- gam(d15N ~ s(Year), data = isotopes, method = 'REML')
```

Family: gaussian

Link function: identity

Formula:

d15N ~ s(Year)

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.30958	0.02805	118	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(Year)	7.466	8.416	70.13	<2e-16 ***

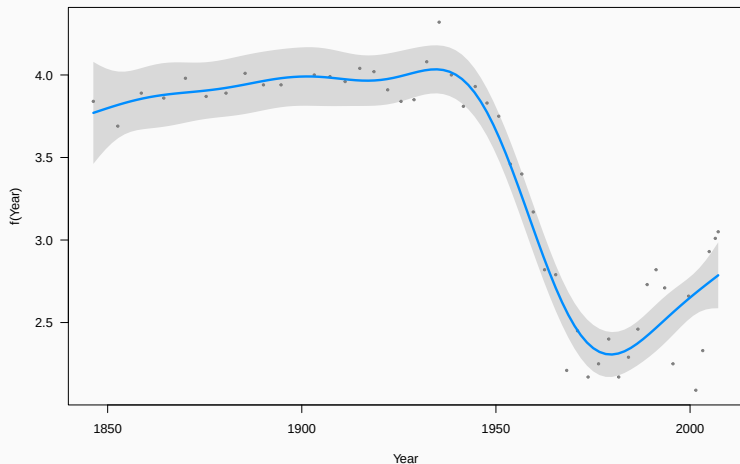
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.926 Deviance explained = 93.8%

-REML = 4.8282 Scale est. = 0.037771 n = 48

Visualising fitted GAM

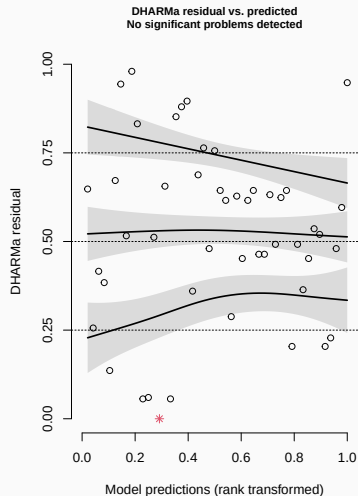
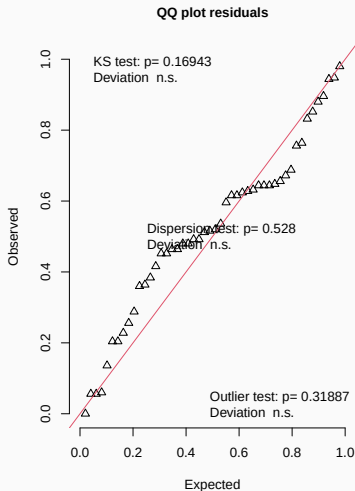
`visreg(m)`



Checking fitted GAM

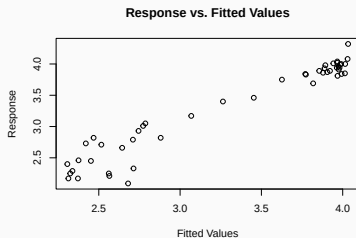
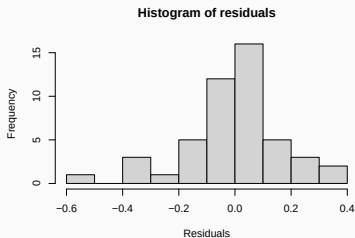
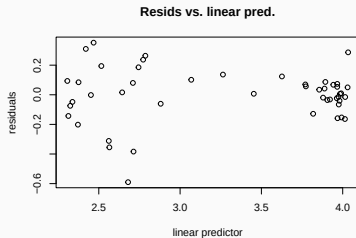
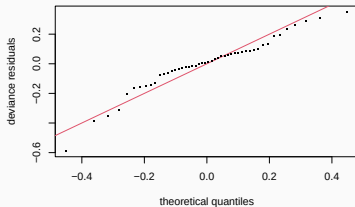
```
library('DHARMa')  
simulateResiduals(m, plot = TRUE)
```

DHARMa residual



Checking fitted GAM

```
gam.check(m)
```



Checking fitted GAM

```
gam.check(m)
```

Method: REML Optimizer: outer newton

full convergence after 6 iterations.

Gradient range [-3.747047e-11,2.175327e-11]

(score 4.828195 & scale 0.03777081).

Hessian positive definite, eigenvalue range [2.252432,23.50383].

Model rank = 10 / 10

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(Year)	9.00	7.47	0.73	0.025 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Increasing k

```
m <- gam(d15N ~ s(Year, k = 12), data = isotopes, method = 'REML')
gam.check(m)
```

```
Method: REML   Optimizer: outer newton
full convergence after 5 iterations.
Gradient range [-1.437177e-06,2.684611e-07]
(score 4.742639 & scale 0.03687964).
Hessian positive definite, eigenvalue range [2.27223,23.61084].
Model rank = 12 / 12
```

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(Year)	11.00	8.12	0.76	0.045 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				

Increasing k

```
m <- gam(d15N ~ s(Year, k = 15), data = isotopes, method = 'REML')
gam.check(m)
```

Method: REML Optimizer: outer newton

full convergence after 5 iterations.

Gradient range [-2.100258e-10,1.486082e-10]

(score 3.973402 & scale 0.0329899).

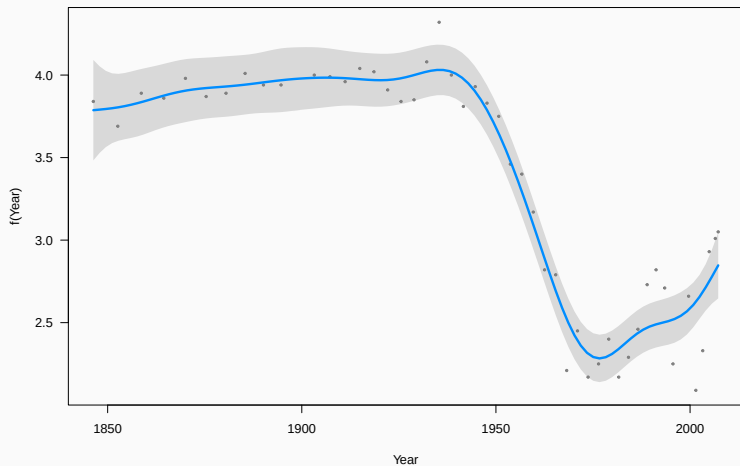
Hessian positive definite, eigenvalue range [1.570402,23.80022].

Model rank = 15 / 15

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

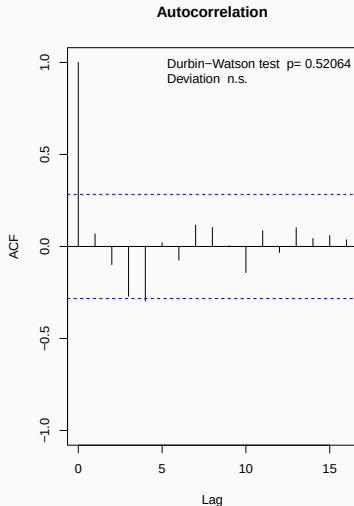
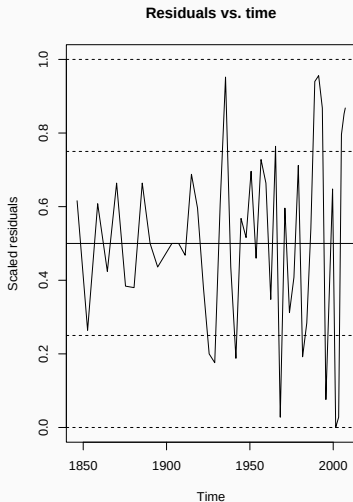
	k'	edf	k-index	p-value
s(Year)	14.00	9.28	0.85	0.11

Visualising fitted GAM



Checking fitted GAM

```
testTemporalAutocorrelation(simulateResiduals(m),  
                             time = isotopes$Year)
```



Including temporal autocorrelation

```
mod <- gamm(d15N ~ s(Year, k = 15), data = isotopes,  
            correlation = corCAR1(form = ~ Year), method = 'REML')
```

Family: gaussian

Link function: identity

Formula:

d15N ~ s(Year, k = 15)

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.30909	0.03489	94.84	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(Year)	7.954	7.954	47.44	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.929

Scale est. = 0.037268 n = 48

Modelling infant mortality

Modelling infant mortality

```
mort <- read.csv('data/UN_GDP_infantmortality.csv')
```

	country	infant.mortality	gdp
1	Afghanistan	154	2848
2	Albania	32	863
3	Algeria	44	1531
4	American.Samoa	11	NA
5	Andorra	NA	NA
6	Angola	124	355

Modelling infant mortality with a GLM

```
library('MASS')  
mort.glm <- glm.nb(infant.mortality ~ gdp, data = mort)
```

Call:

```
glm.nb(formula = infant.mortality ~ gdp, data = mort, init.theta = 2.460991808,  
       link = log)
```

Coefficients:

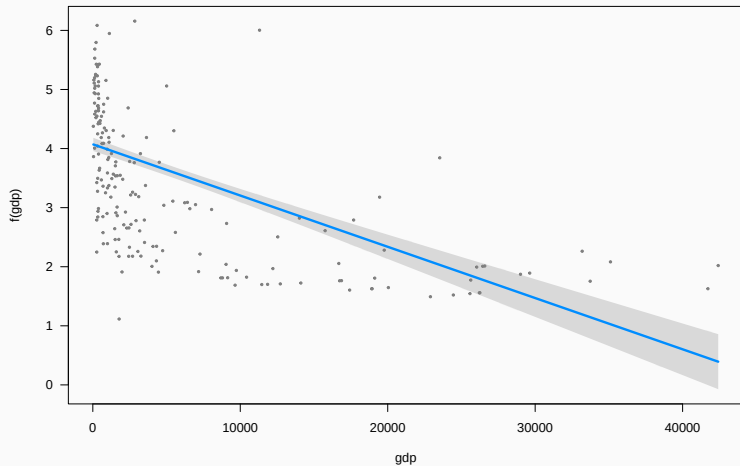
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	4.072e+00	5.727e-02	71.11	<2e-16 ***
gdp	-8.675e-05	6.221e-06	-13.95	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(2.461) family taken to be 1)

Null deviance: 385.83 on 192 degrees of freedom
Residual deviance: 202.51 on 191 degrees of freedom
(14 observations deleted due to missingness)
AIC: 1715

Modelling infant mortality with a GLM



Modelling infant mortality with a GLM (log.gdp)

```
mort$log.gdp <- log(mort$gdp)
mort.glm.log <- glm.nb(infant.mortality ~ log.gdp, data = mort)
```

Call:

```
glm.nb(formula = infant.mortality ~ log.gdp, data = mort, init.theta = 3.119314453,
       link = log)
```

Coefficients:

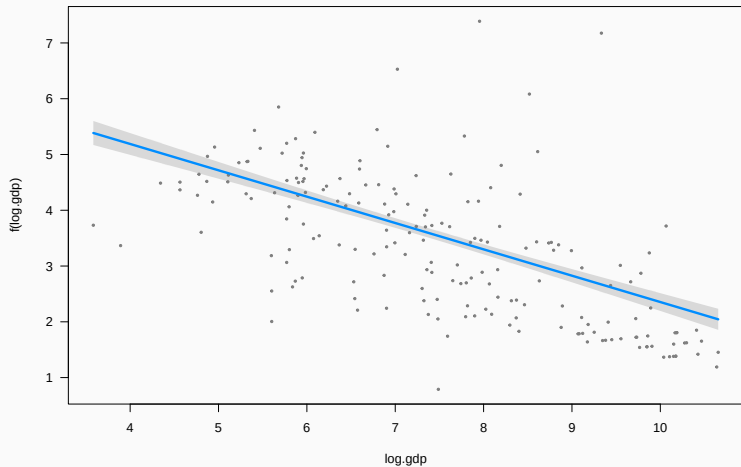
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	7.07818	0.20045	35.31	<2e-16 ***
log.gdp	-0.47238	0.02647	-17.85	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(3.1193) family taken to be 1)

Null deviance: 478.54 on 192 degrees of freedom
Residual deviance: 198.03 on 191 degrees of freedom
(14 observations deleted due to missingness)
AIC: 1667.7

Modelling infant mortality with a GLM (log.gdp)



Modelling infant mortality with a GAM

```
library('mgcv')  
mort.gam <- gam(infant.mortality ~ s(log.gdp), family = nb, data = mort)
```

Family: Negative Binomial(3.251)

Link function: log

Formula:

infant.mortality ~ s(log.gdp)

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.51137	0.04257	82.49	<2e-16 ***

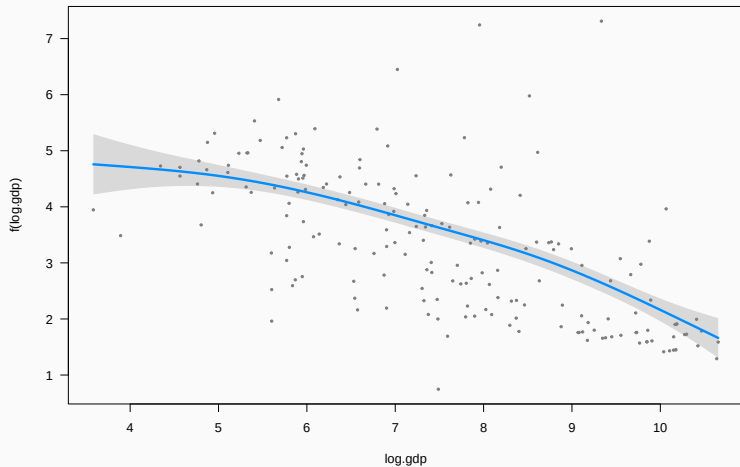
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(log.gdp)	3.134	3.937	329.9	<2e-16 ***

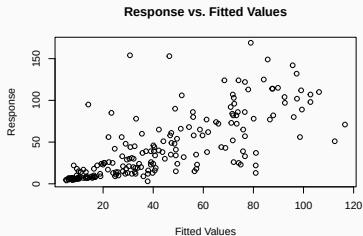
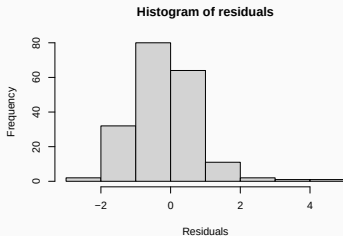
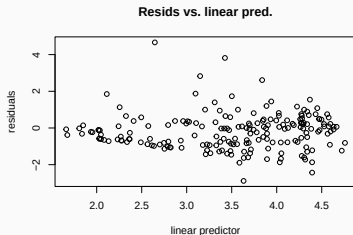
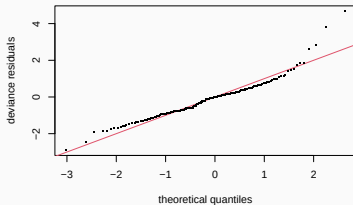
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Modelling infant mortality with a GAM



Checking GAM

```
gam.check(mort.gam)
```



Checking GAM

```
gam.check(mort.gam)
```

Method: REML Optimizer: outer newton

full convergence after 4 iterations.

Gradient range [4.395199e-08,2.344699e-06]

(score 832.1006 & scale 1).

Hessian positive definite, eigenvalue range [0.4630879,82.57877].

Model rank = 10 / 10

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(log.gdp)	9.00	3.13	0.92	0.24

Comparing models

```
library('easystats')  
compare_performance(mort.glm, mort.glm.log, mort.gam)
```

Comparison of Model Performance Indices

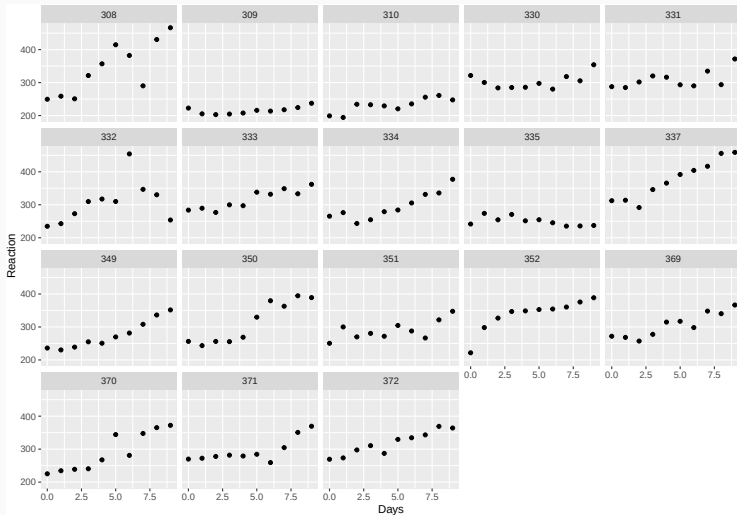
Name	Model	AIC (weights)	AICc (weights)	BIC (weights)
mort.glm	negbin	1715.0 (<.001)	1715.1 (<.001)	1724.7 (<.001)
mort.glm.log	negbin	1667.7 (0.035)	1667.9 (0.041)	1677.5 (0.816)
mort.gam	gam	1661.1 (0.965)	1661.6 (0.959)	1680.5 (0.184)

Name	RMSE	Sigma	Score_log	Score_spherical	Nagelkerke's R2	R2
mort.glm	31.089	1	-4.437	0.048	0.709	
mort.glm.log	30.034	1	-4.356	0.053	0.836	
mort.gam	26.249	1	-4.296	0.049		0.526

Generalised Additive Mixed Models (GAMM)

Reaction time with sleep deprivation

```
library('lme4')  
data('sleepstudy')
```



Modelling reaction time with sleep deprivation (GAMM)

```
sgamm <- gam(Reaction ~ s(Days) + s(Subject, bs = "re"), data = sleepstudy, method = 'REML')
```

Family: gaussian

Link function: identity

Formula:

Reaction ~ s(Days) + s(Subject, bs = "re")

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	298.51	9.05	32.98	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(Days)	1.006	1.011	167.65	<2e-16 ***
s(Subject)	15.892	17.000	14.35	<2e-16 ***

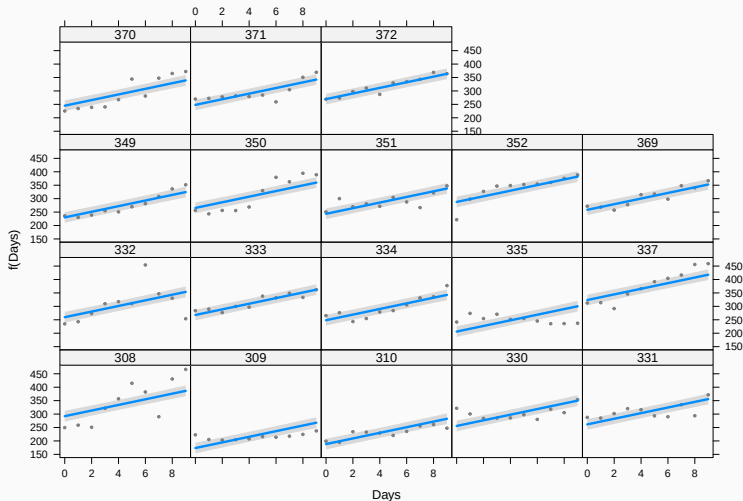
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.697 Deviance explained = 72.6%

-REML = 892.18 Scale est. = 960.43 n = 180

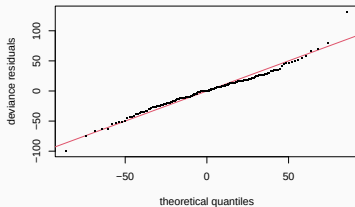
Modelling reaction time with sleep deprivation (GAMM)

```
visreg(sgamma, xvar = 'Days', by = 'Subject')
```

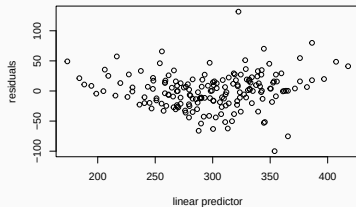


Modelling reaction time with sleep deprivation (GAMM)

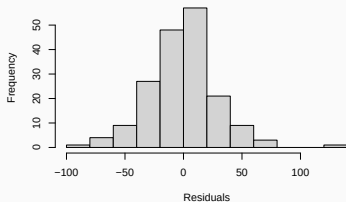
```
gam.check(sgamm)
```



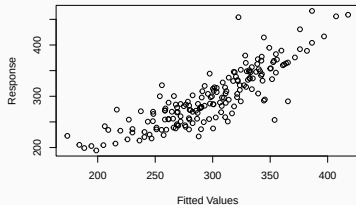
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



Modelling reaction time with sleep deprivation (GAMM)

```
gam.check(sgamm)
```

```
Method: REML    Optimizer: outer newton  
full convergence after 7 iterations.  
Gradient range [-0.0003899837,0.002002328]  
(score 892.1778 & scale 960.4321).  
Hessian positive definite, eigenvalue range [0.0004006477,89.76488].  
Model rank = 28 / 28
```

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(Days)	9.00	1.01	1.07	0.81
s(Subject)	18.00	15.89	NA	NA

Modelling reaction time with sleep deprivation (GAMM)

```
sgamm <- gam(Reaction ~ s(Days, Subject, k = 3, bs = 'fs'),  
             data = sleepstudy, method = 'REML')
```

Family: gaussian

Link function: identity

Formula:

Reaction ~ s(Days, Subject, k = 3, bs = "fs")

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	298.51	9.05	32.98	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(Days,Subject)	45.67	53	17.11	<2e-16 ***

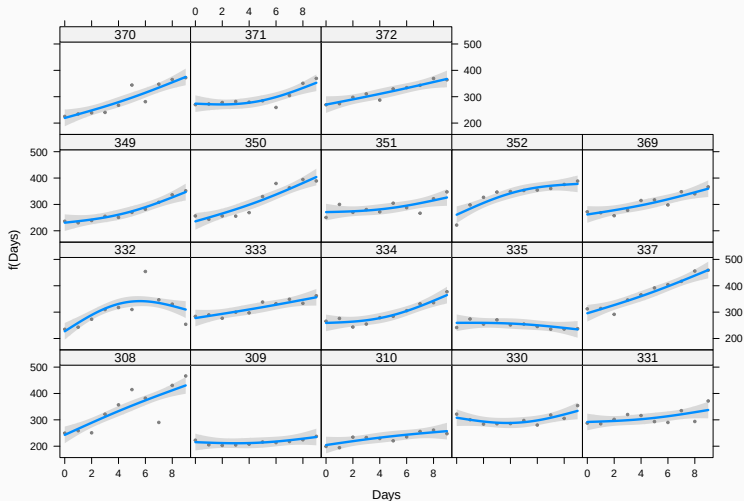
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) = 0.835 Deviance explained = 87.7%

-REML = 883.64 Scale est. = 523.2 n = 180

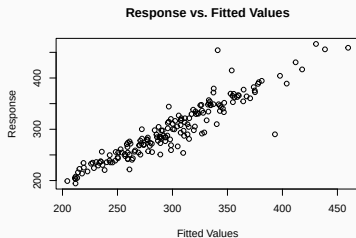
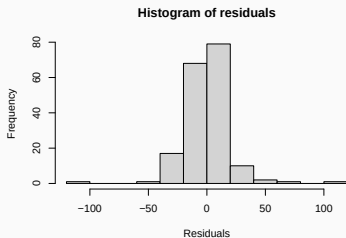
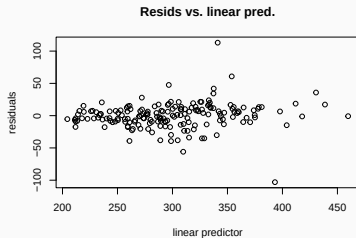
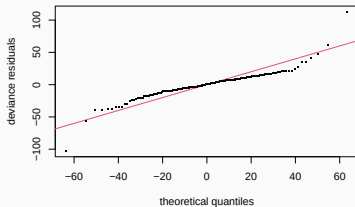
Modelling reaction time with sleep deprivation (GAMM)

```
visreg(sgamma, xvar = 'Days', by = 'Subject')
```



Modelling reaction time with sleep deprivation (GAMM)

```
gam.check(sgamm)
```



Modelling reaction time with sleep deprivation (GAMM)

```
gam.check(sgam)
```

Method: REML Optimizer: outer newton

full convergence after 6 iterations.

Gradient range [-8.770542e-09,3.944987e-09]

(score 883.6432 & scale 523.1968).

Hessian positive definite, eigenvalue range [3.37416,91.60746].

Model rank = 55 / 55

Basis dimension (k) checking results. Low p-value (k-index<1) may indicate that k is too low, especially if edf is close to k'.

	k'	edf	k-index	p-value
s(Days,Subject)	54.0	45.7	1	0.47

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